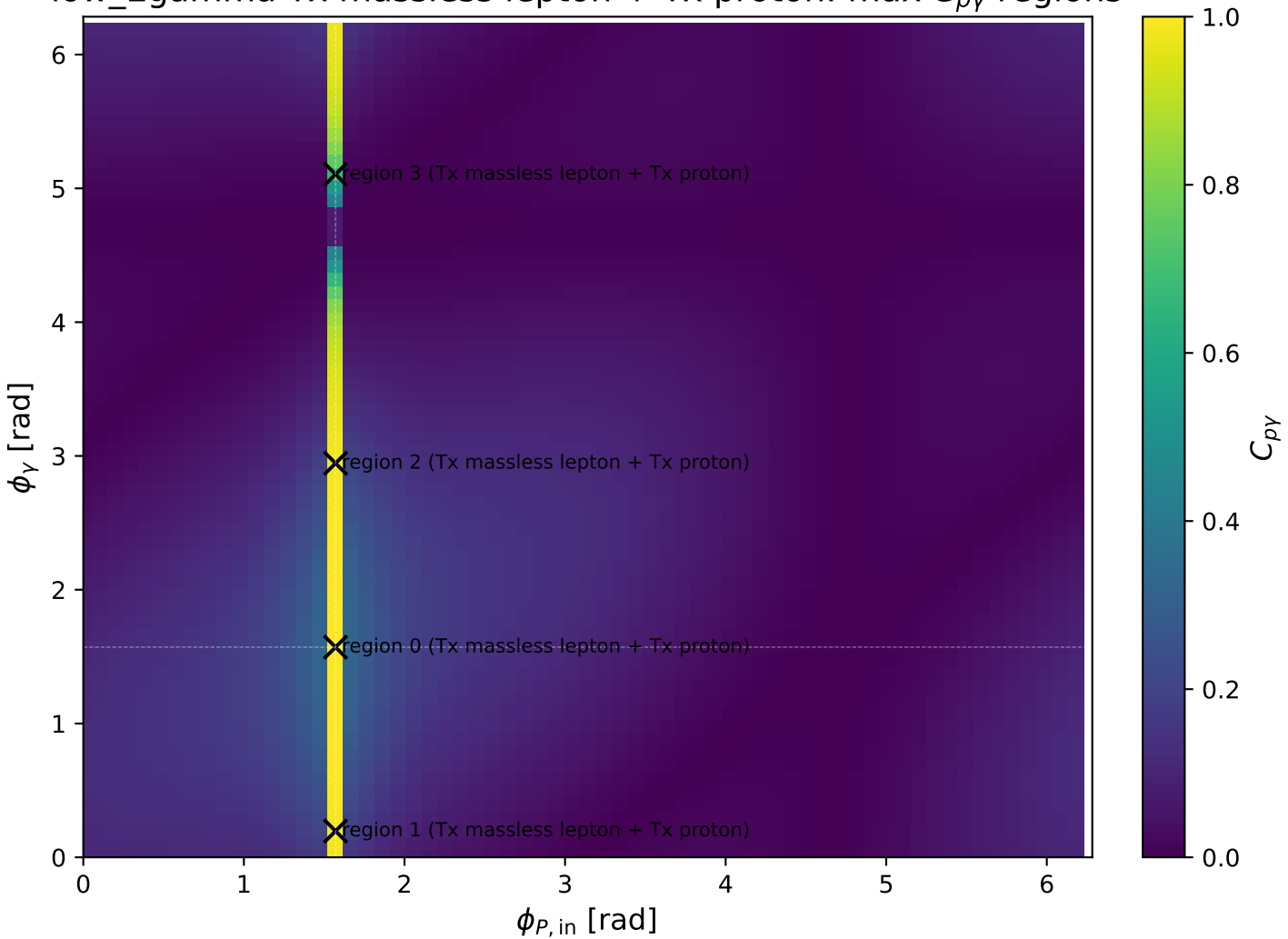
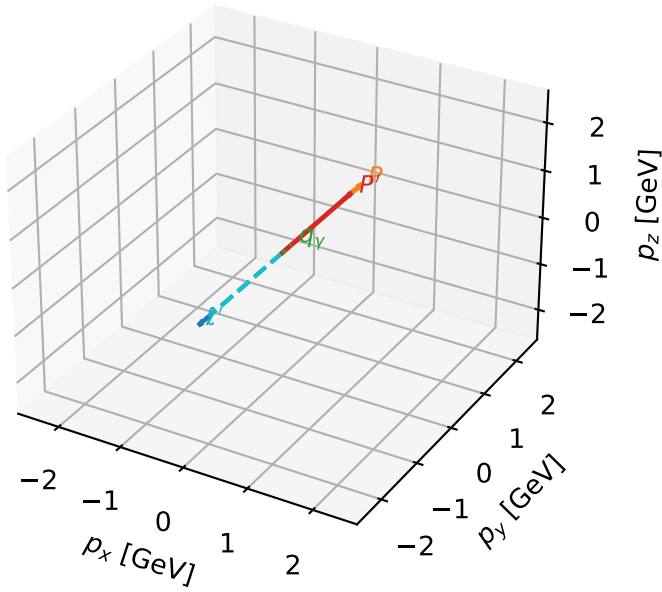


low\_Egamma Tx massless lepton + Tx proton: max  $C_{py}$  regions



# Tx massless lepton + Tx proton characteristic max $C_{p\gamma}$ configuration

3D momenta

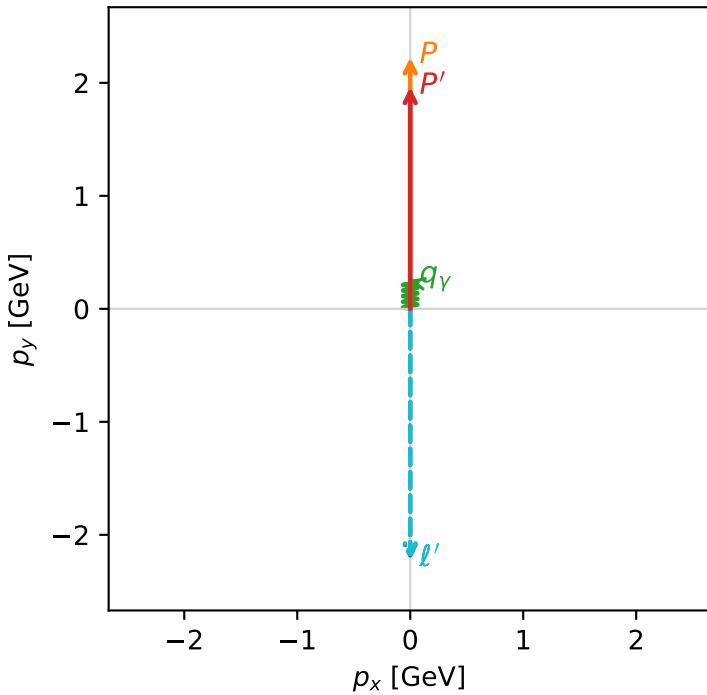


c\_p\_gamma\_low\_egamma\_txtx\_region\_0 C\_{p\gamma}=0.999987  
 region: low\_Egamma TxTx\_region\_0 final-pair delta\_xy=0 rad  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.96296$   
 $\theta_{in}=1.57079$ ,  $\phi_{\ell}=4.71239$ ,  $\phi_p=1.5708$   
 $E_{\gamma}=0.25$ ,  $\phi_{\gamma}=1.5708$   
 $Q^2=8.66557e-12$ ,  $x_B=8.15204e-11$ ,  $t=-0.0114583$ ,  $W^2=0.986143$ ,  $y=0.00515116$   
 $\theta(\ell', \gamma)=180$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.2130$   $p_x= -0.0000$   $p_y= -2.2130$   $p_z= 0.0000$   
 $P'$   $E= 2.1756$   $p_x= 0.0000$   $p_y= 1.9630$   $p_z= 0.0000$   
 $q_{\gamma}$   $E= 0.2500$   $p_x= 0.0000$   $p_y= 0.2500$   $p_z= 0.0000$

Transverse plane



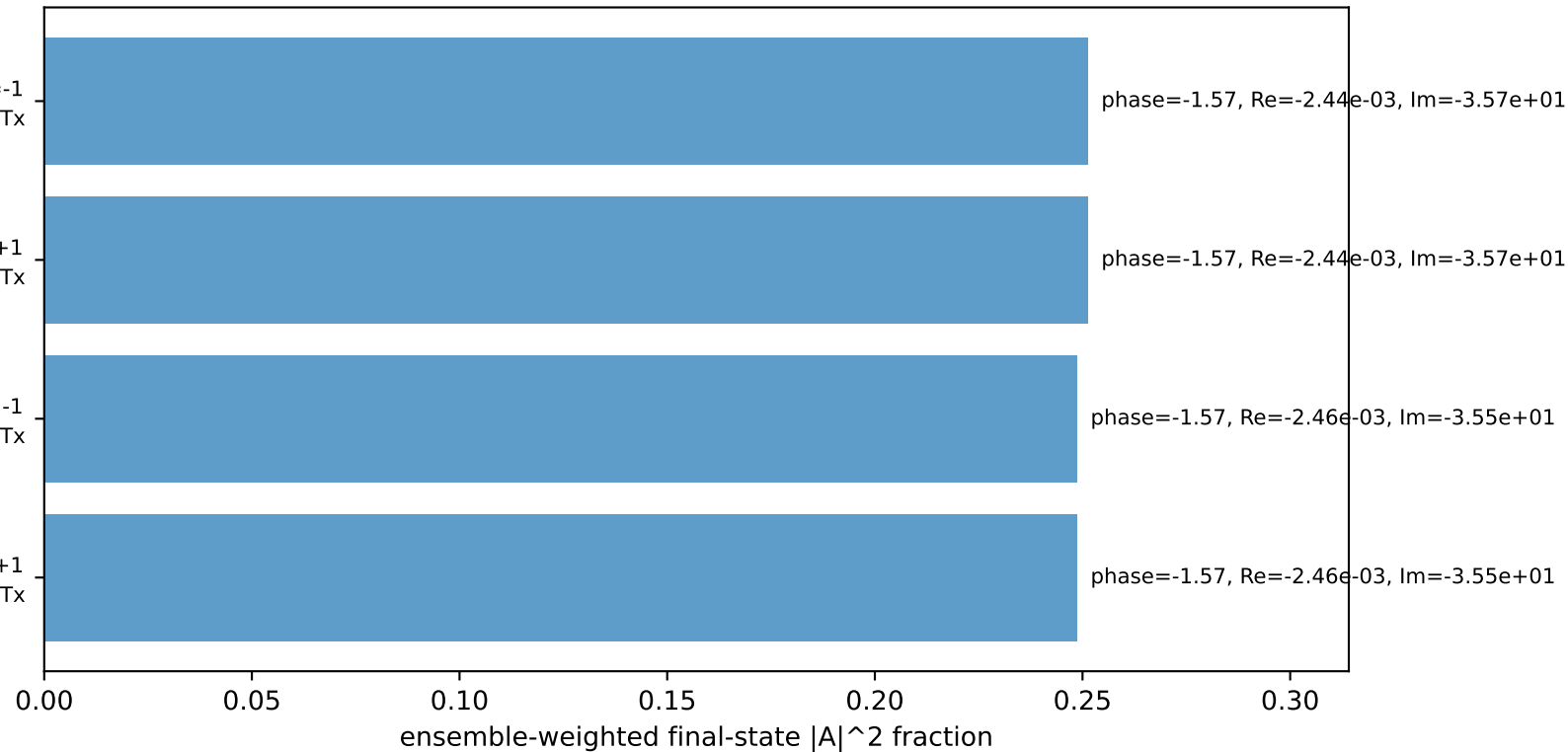
$h_e=-1, h_p=+1, h_{\gamma}=-1$   
 electron Tx, proton Tx

$h_e=+1, h_p=-1, h_{\gamma}=+1$   
 electron Tx, proton Tx

$h_e=+1, h_p=+1, h_{\gamma}=-1$   
 electron Tx, proton Tx

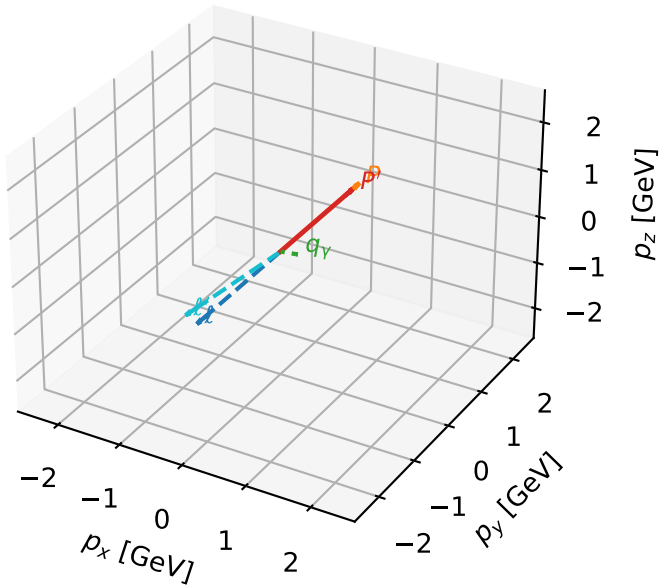
$h_e=-1, h_p=-1, h_{\gamma}=+1$   
 electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=4, retained=100.0%)



# Tx massless lepton + Tx proton characteristic max $C_{p\gamma}$ configuration

3D momenta

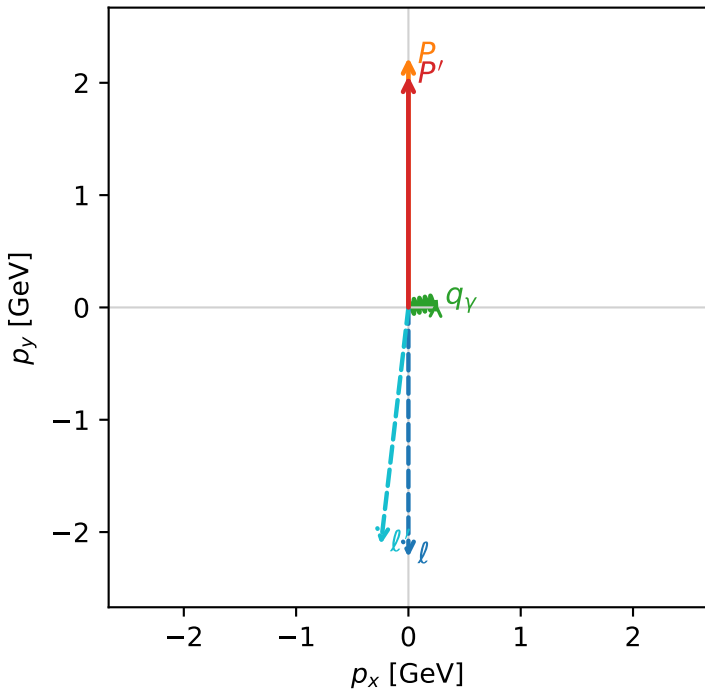


c\_p\_gamma\_low\_egamma\_txtx\_region\_1 C\_{p\gamma}=0.984344  
 region: low\_Egamma TxTx\_region\_1 final-pair delta\_xy=1.37445 rad  
 outgoing-state purity: 1  
 kinematic point: high\_s high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,l_0}\rangle \otimes |T_{x,p}\rangle$

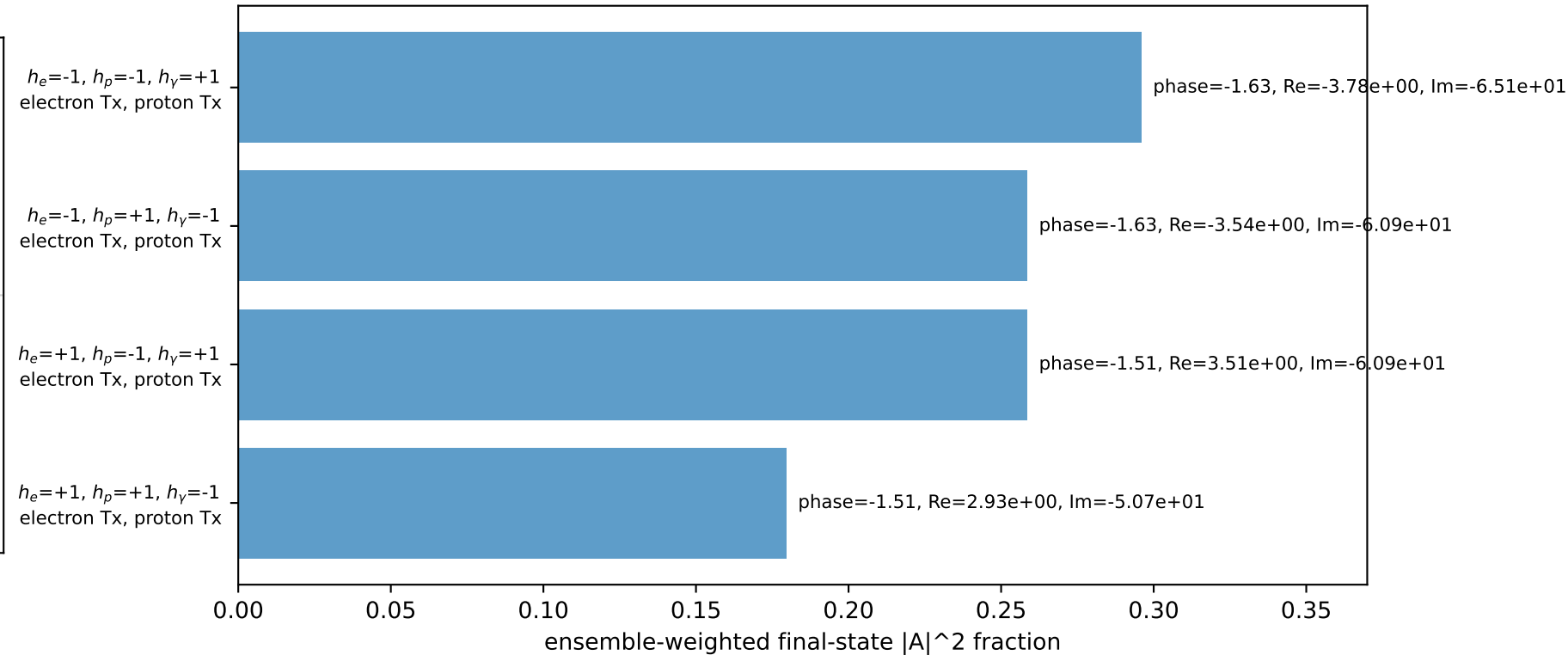
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.06107$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=0.19635$   
 $Q^2=0.0631736$ ,  $x_B=0.0635316$ ,  $t=-0.00429535$ ,  $W^2=1.81103$ ,  $y=0.0481859$   
 $\theta(l', \gamma)=107.879$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $l'$   $E= 2.1240$   $p_x= -0.2452$   $p_y= -2.1098$   $p_z= 0.0000$   
 $P'$   $E= 2.2645$   $p_x= 0.0000$   $p_y= 2.0611$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.2452$   $p_y= 0.0488$   $p_z= 0.0000$

Transverse plane

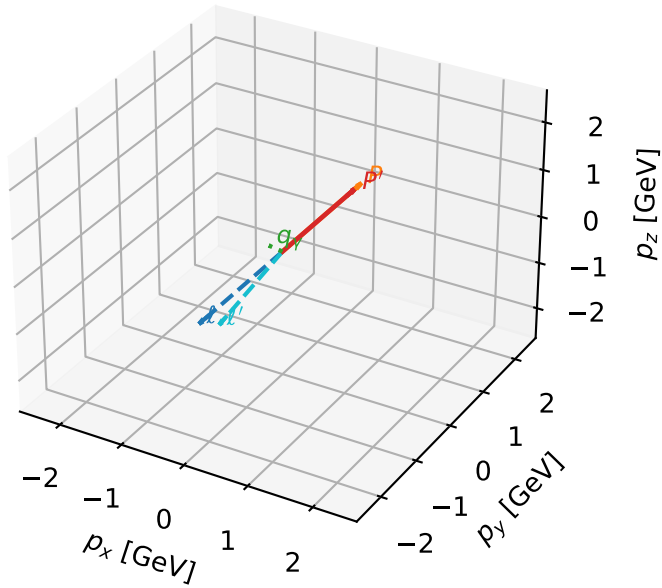


Leading initial-ensemble/final-state components (N=4, retained=99.3%)



# Tx massless lepton + Tx proton characteristic max $C_{p\gamma}$ configuration

3D momenta



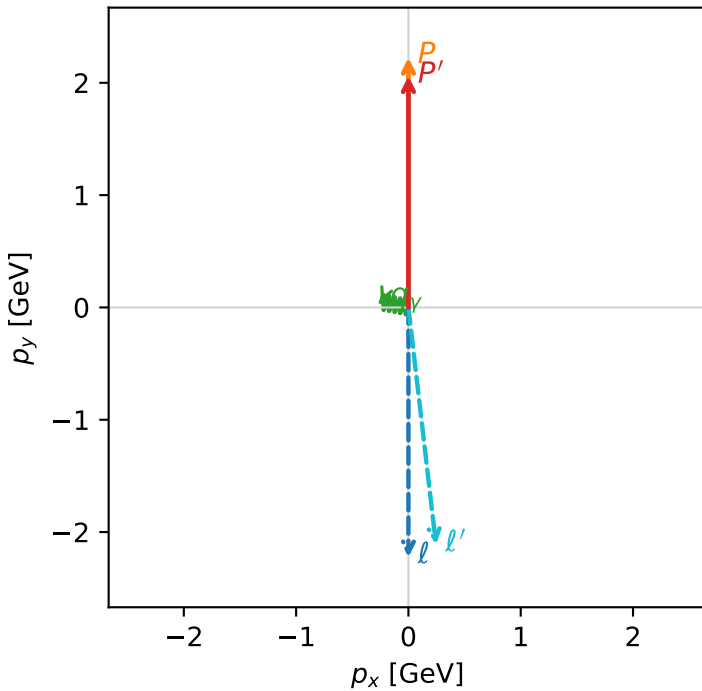
c\_p\_gamma\_low\_egamma\_txtx\_region\_2 C\_{p\gamma}=0.984344  
 region: low\_Egamma TxTx\_region\_2 final-pair delta\_xy=1.37445 rad  
 outgoing-state purity: 1  
 kinematic point: high\_s high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.06107$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=2.94524$   
 $Q^2=0.0631736$ ,  $x_B=0.0635316$ ,  $t=-0.00429535$ ,  $W^2=1.81103$ ,  $y=0.0481859$   
 $\theta(\ell', \gamma)=107.879$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1240$   $p_x= 0.2452$   $p_y= -2.1098$   $p_z= 0.0000$   
 $P'$   $E= 2.2645$   $p_x= 0.0000$   $p_y= 2.0611$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= -0.2452$   $p_y= 0.0488$   $p_z= 0.0000$

Transverse plane



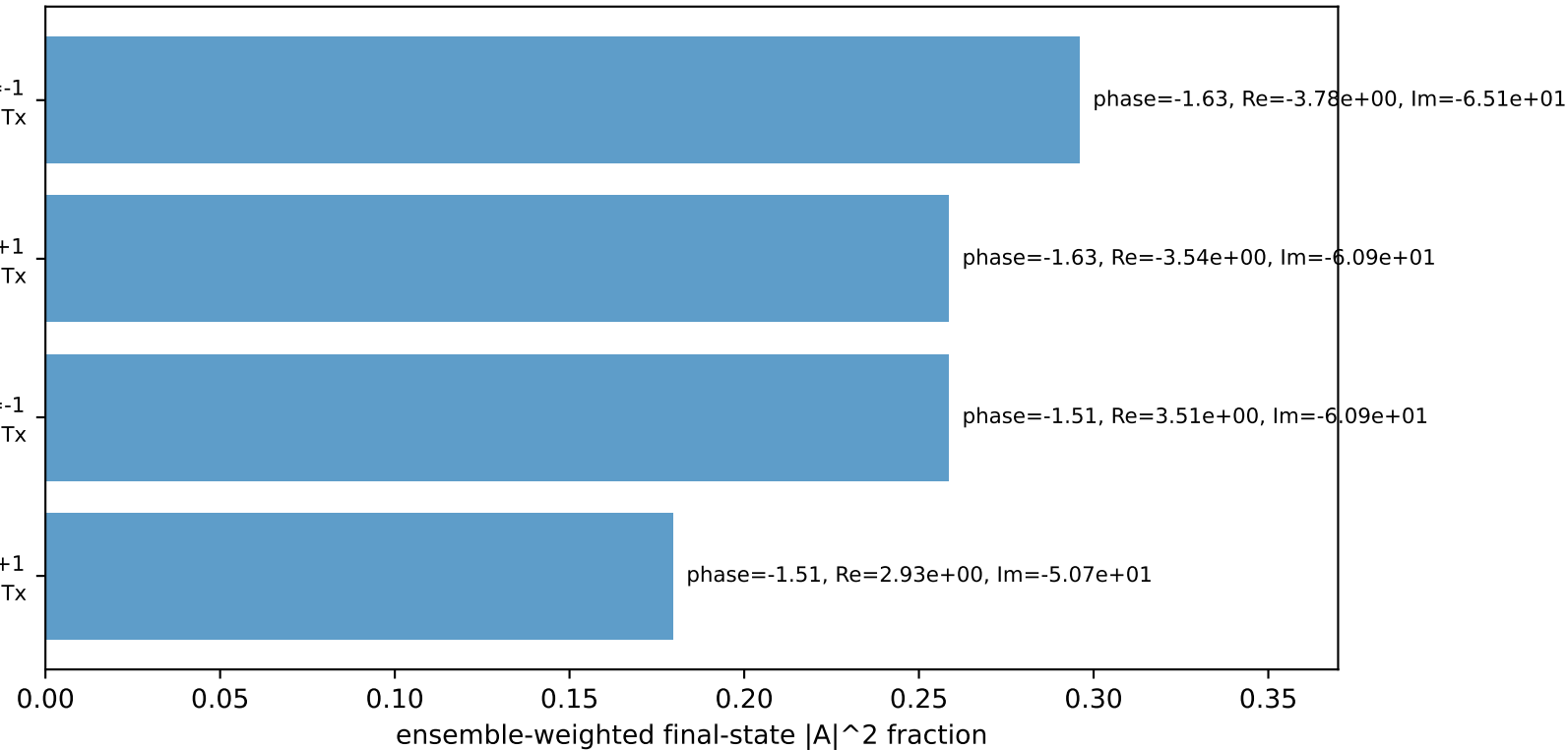
$h_e=+1, h_p=+1, h_\gamma=-1$   
 electron Tx, proton Tx

$h_e=+1, h_p=-1, h_\gamma=+1$   
 electron Tx, proton Tx

$h_e=-1, h_p=+1, h_\gamma=-1$   
 electron Tx, proton Tx

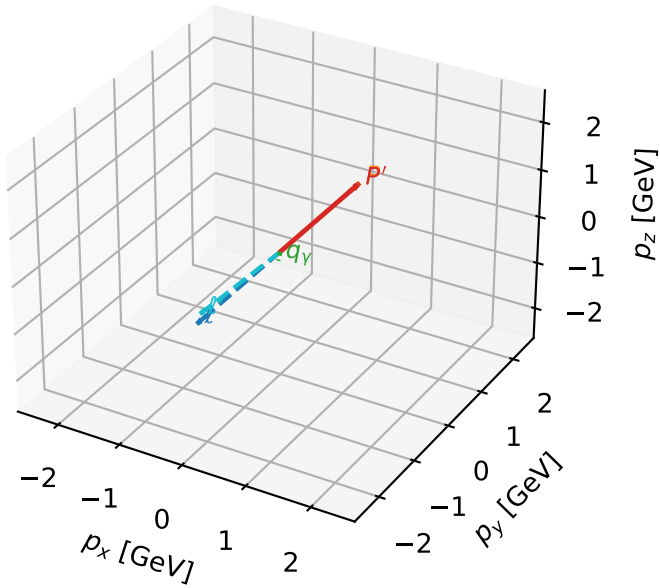
$h_e=-1, h_p=-1, h_\gamma=+1$   
 electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=4, retained=99.3%)



# Tx massless lepton + Tx proton characteristic max $C_{p\gamma}$ configuration

3D momenta



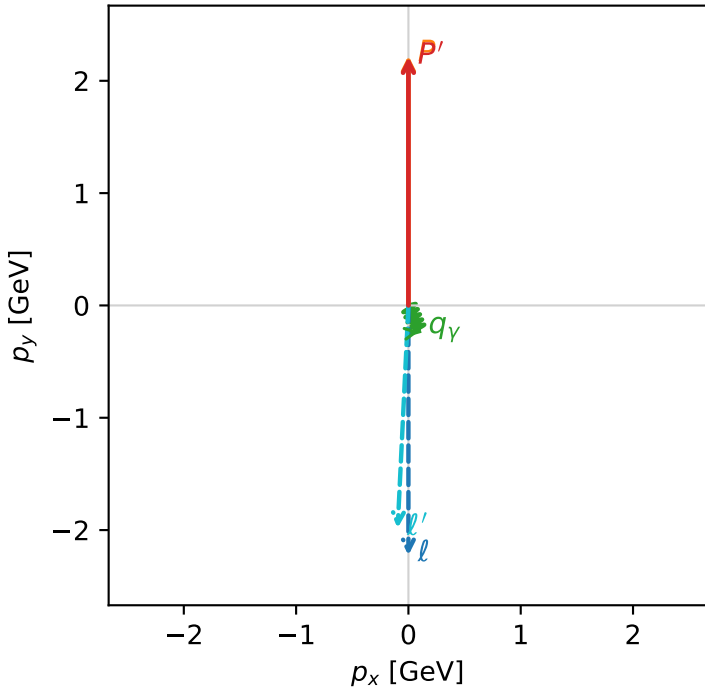
c\_p\_gamma\_low\_egamma\_txtx\_region\_3 C\_{p\gamma}=0.658932  
 region: low\_Egamma TxTx\_region\_3 final-pair delta\_xy=2.74889 rad  
 outgoing-state purity: 1  
 kinematic point: high\_s high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.21331$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=5.10509$   
 $Q^2=0.0102647$ ,  $x_B=0.00459348$ ,  $t=-1.87039e-05$ ,  $W^2=3.10419$ ,  $y=0.108287$   
 $\theta(\ell', \gamma)=25.263$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 1.9846$   $p_x= -0.0957$   $p_y= -1.9823$   $p_z= 0.0000$   
 $P'$   $E= 2.4039$   $p_x= 0.0000$   $p_y= 2.2133$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.0957$   $p_y= -0.2310$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=5, retained=99.0%)

